

Smart Clinic Database System

Mid-Project Progress Report

Student Name	Student ID	Primary Contribution
Dana Mohammed Alajmi	240040990	Group Leader and Repository Manager
Amani Barjas	240047827	Task 1: Database Design
Dareen Alanazi	240022387	Task 2: Database Implementation
Manar Mutaib Al-Harbi	240061219	Task 3: SQL Operations

1. Progress Summary

The project has completed the Task 3 preparation stage. The single MySQL file now contains the required SELECT, JOIN, nested, GROUP BY, UPDATE, DELETE, VIEW, and TRIGGER operations with purpose statements. The file passed MySQL parsing and internal data checks, and equivalent operations returned coherent results in an isolated Supabase PostgreSQL validation schema. Actual MySQL Workbench execution and course-compliant screenshot capture remain required before final approval.

Workstream	Status	Evidence/Output
Task 1 - Design	Prepared for team review	ER/EER diagram, relational schema, keys, cardinalities, assumptions
Task 2 - Implementation	Prepared for MySQL execution	Database, eight tables, constraints, and sample records
Task 2 Evidence	Pending actual capture	MySQL Workbench code, table outputs, and execution status
Task 3 - SQL Operations	Implemented and support-validated	MySQL code plus eight genuine, clearly labeled Supabase result panels
Task 3 MySQL Evidence	Pending actual capture	MySQL Workbench code and output screenshots for every assessed operation
Task 4 - Reflection	Deferred by instruction	Will be written only after the full project is complete

2. Completed Work

1. Defined eight relational tables covering people, patients, doctors, appointments, treatments, medicines, prescriptions, and payments.
2. Modeled PERSON as a supertype specialized into the PATIENT and DOCTOR subtypes.
3. Specified primary keys, foreign keys, uniqueness constraints, checks, and controlled ENUM values.
4. Produced a complete ER/EER diagram with entity attributes and relationship cardinalities.
5. Prepared a repeatable MySQL 8.0 script with a transaction-based insertion sequence.
6. Inserted at least five logically connected records into every main table.
7. Added SELECT, JOIN, nested, GROUP BY, UPDATE, DELETE, VIEW, and TRIGGER operations to the same SQL file.
8. Executed equivalent Task 3 operations in an isolated Supabase schema and restored all temporary data changes.
9. Created eight genuine evidence panels that identify their platform and do not impersonate MySQL Workbench.

3. Key Design Decisions

Decision	Reason
Shared person supertype	Avoids repeated identity and contact attributes in separate patient and doctor tables.
Disjoint, total specialization	Makes each clinic person one clear role while demonstrating the required EER feature.
Associative prescription table	Stores dosage and duration while resolving the treatment-medicine many-to-many relationship.
Multiple payments per appointment	Supports insurance, installments, and future mixed payment scenarios.
Restrictive transactional deletion	Protects clinical and financial history from accidental parent-record deletion.
One evolving SQL file	Meets the final requirement for a single implementation file and prevents inconsistent copies.

4. Challenges and Responses

Design-Stage Challenge	Response
Representing specialization relationally	Used PERSON as the supertype and made each subtype key a foreign key to PERSON.
Maintaining referential insertion order	Created and populated tables from parent entities to dependent transactional entities inside one transaction.
Preventing schedule conflicts	Added unique doctor-time and patient-time constraints to APPOINTMENT.
Protecting evidence integrity	Kept MySQL placeholders, added visibly labeled supporting Supabase results, and avoided presenting them as Workbench evidence.
Trigger portability	Qualified the PostgreSQL validation function's schema, set an immutable search path, and retained the correct MySQL trigger syntax in the assessed file.

5. Team Contributions

Member	Current Contribution	Status
Dana Mohammed Alajmi	Project coordination, file organization, repository management, integration review	In progress
Amani Barjas	Task 1 entity analysis, keys, relationships, cardinalities, EER feature, assumptions, relational schema	Prepared for review
Dareen Alanazi	Task 2 database creation, table definitions, constraints, data insertion, and testing preparation	Prepared for MySQL execution
Manar Mutaib Al-Harbi	Task 3 SELECT, JOIN, nested, aggregate, UPDATE, DELETE, VIEW, TRIGGER, purposes, and evidence specification	Prepared for team review
All members	Cross-review, testing, evidence, progress report, final report, and later reflection	Ongoing

6. Next-Stage Plan

1. We will review the ER/EER model, assumptions, relational schema, sample data, and Task 3 purposes as a team.
2. We will run the complete Tasks 1–3 script in MySQL Workbench and correct any environment-specific issues.
3. We will capture and organize all required Task 2 and Task 3 MySQL Workbench screenshots.
4. We will compare the MySQL results with the supporting validation evidence and approve the final technical results.
5. We will review the actual MySQL query results before writing the 300–500-word Project Reflection.
6. We will assemble and verify the final DOCX, PDF, correct repository link, and compressed submission package.